ECSE 331
Laboratory No. 1
Introduction to PSPICE[1]

Objective

To get familiarized with circuit simulation using PSPICE.

Introduction

SPICE (Simulation Program for Integrated Circuits Emphasis) is a general purpose circuit simulator. It was originally developed at the Electronics Research Laboratory of the University of California, Berkeley (1975). PSpice is a PC version of SPICE which is currently available from OrCAD Corp. of Cadence Design Systems, Inc.

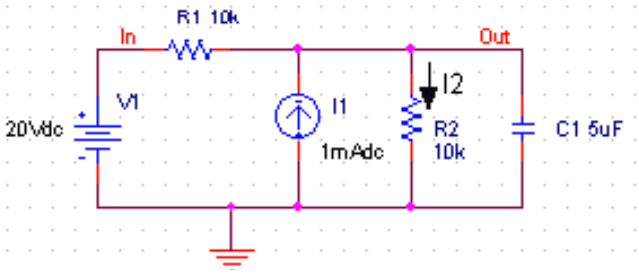
Practical Information for the Student

PSPICE Student version 9.1 (for Windows OS only) is available for free download from the following link: https://www.electronics-lab.com/downloads/circutedesignsimulation/?page=5

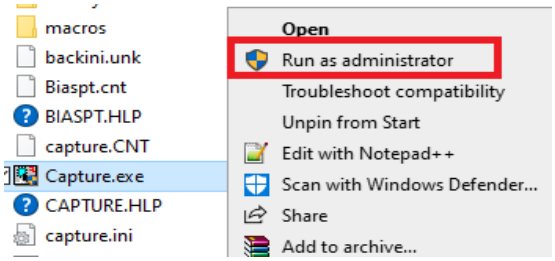
It is important to follow the installation guide in the link below to ensure that the software runs smoothly: https://ece.northeastern.edu/courses/eece2412/2016fa/Windows_10_-_PSpice_Installation_Guide.pdf

Demonstration of simulation using PSPICE

We shall demonstrate the basics of simulation using PSPICE using the following circuit as example:



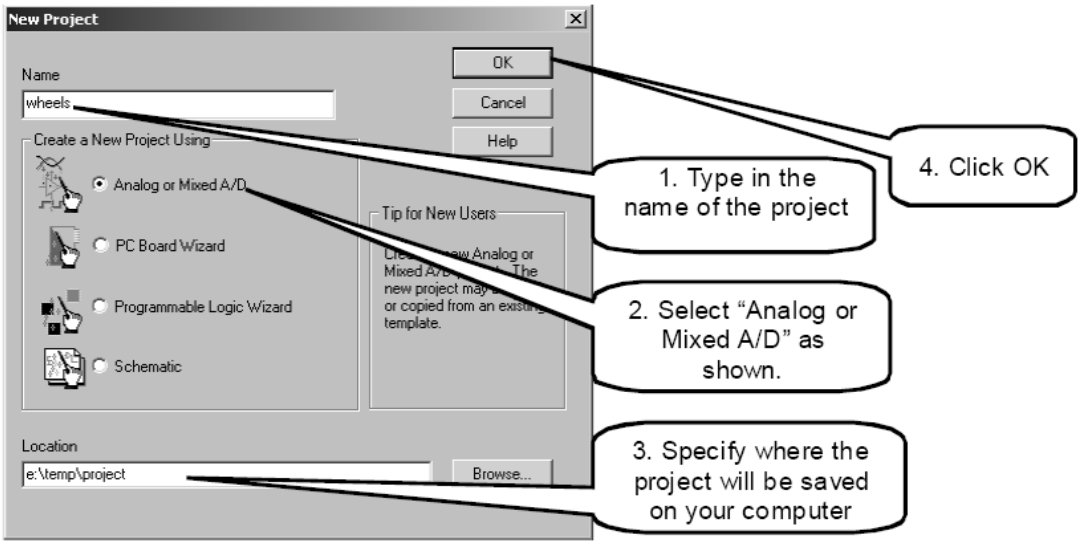
1. Run Capture.exe ***as administrator.***



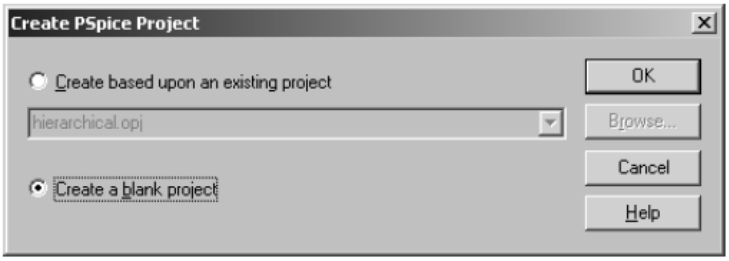
2. From the File menu in Capture choose New → Project.

3. The following dialog will appear. Fill in the information as shown.

1 The choice of PSPICE as the simulation software is not strict. The student may use any other simulation tool he/she is comfortable with.

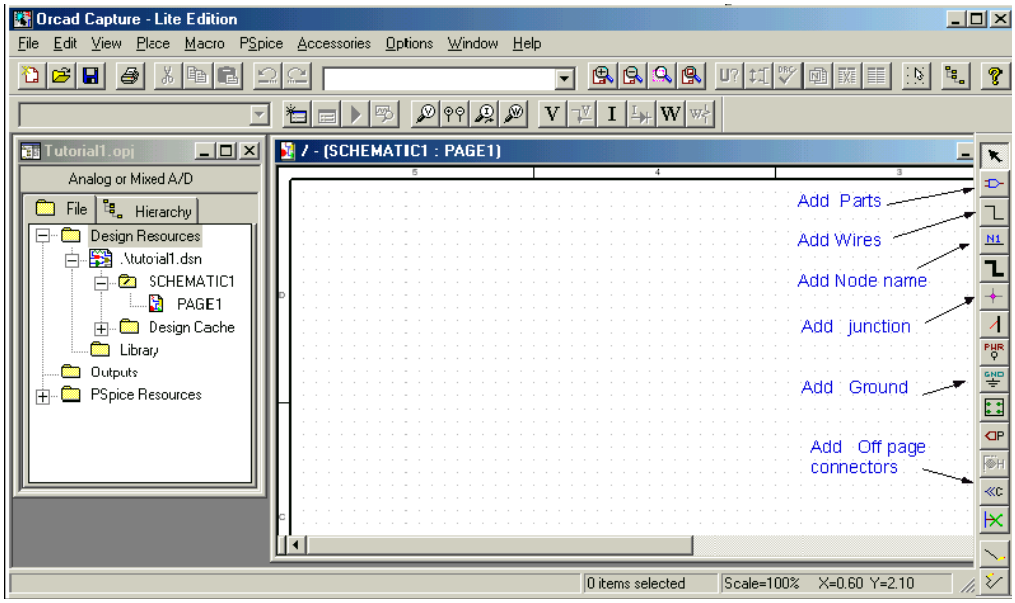


4. The following dialog will appear.

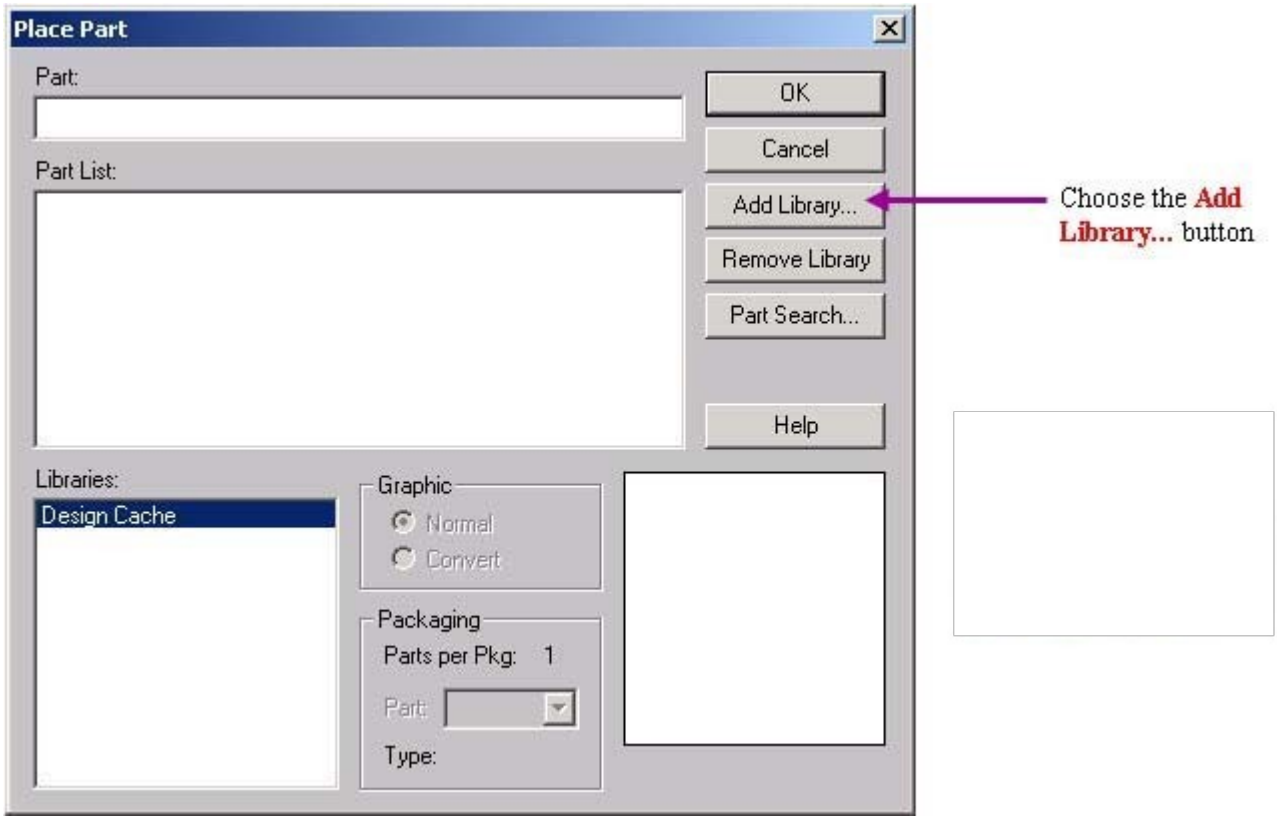


Choose "Create a blank project" and click OK.

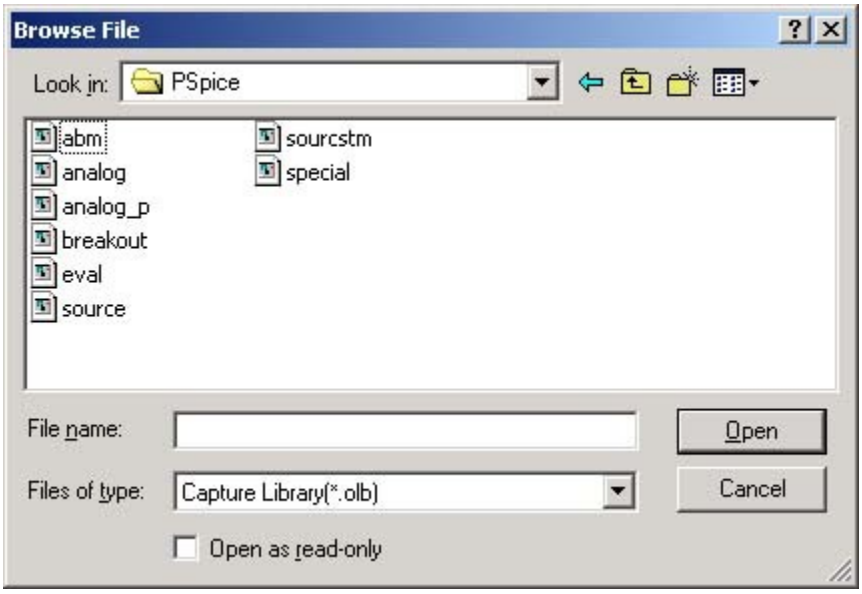
5. The schematic editor will open.



6. Click on the Add Parts Icon. This will open the dialog box shown below.



7. If you are using PSPICE for the first time on your computer, only “Design Cache” is available under libraries as shown above. Click on Add Library. Go to: C -> Program Files -> OrCAD_Demo -> Capture -> Library -> Pspice, After you have reached this location, the Browse File dialog box should look like the one below.



8. Select all the available libraries and click Open. Then the libraries will be added to the Place Part window. A brief description of the libraries is as follows:

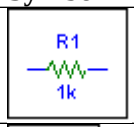
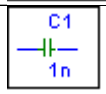
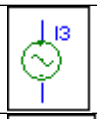
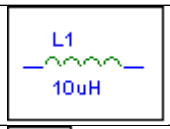
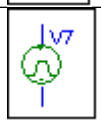
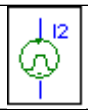
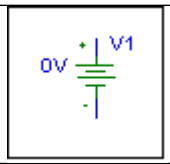
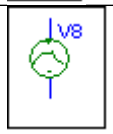
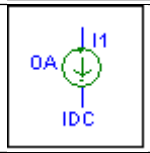
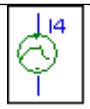
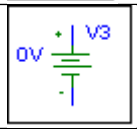
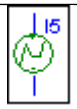
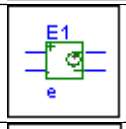
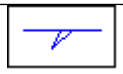
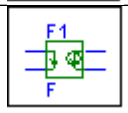
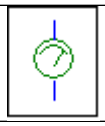
Analog: contains the passive components (R,L,C), mutual inductane, transmission line, and voltage and current dependent sources (voltage dependent voltage source E, current-dependent current source F, voltage-dependent current source G and current-dependent voltage source H).

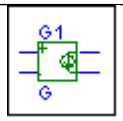
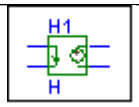
Source: give the different type of independent voltage and current sources, such as Vdc, Idc, Vac, Iac, Vsin, Vexp, pulse, piecewise linear, etc. Browse the library to see what is available.

Eval: provides diodes (D…), bipolar transistors (Q…), MOS transistors, JFETs (J…), real opamp such as the u741, switches (SW_tClose, SW_tOpen), various digital gates and components.

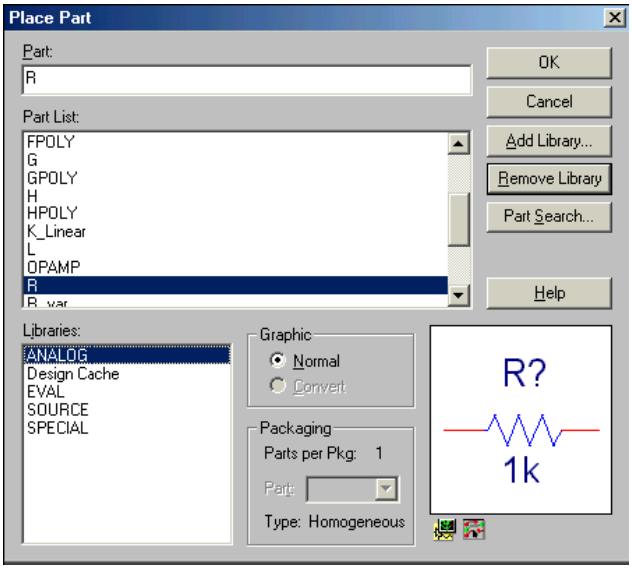
Abm: contains a selection of interesting mathematical operators that can be applied to signals, such as multiplication (MULT), summation (SUM), Square Root (SWRT), Laplace (LAPLACE), arctan (ARCTAN), and many more.

Special: contains a variety of other components, such as PARAM, NODESET, etc.

Part	Name	Symbol	Part	Name	Symbol
Resistor	R		Sinusoidal Voltage Source	VSIN	
Capacitor	C		Sinusoidal Current Source	ISIN	
Inductor	L		Voltage Pulse	VPULSE	
Ground	AGND		Current Pulse	IPULSE	
DC Voltage Source	VDC		Voltage exponential source	VEXP	
DC Current Source	IDC		Current exponential source	IEXP	
AC Voltage Source	VAC		Voltage piecewise linear source	VPWL	
AC Current Source	VDC		Current piecewise linear source	IPWL	
VCVS Voltage-Controlled Voltage Source	E		Voltage Viewpoint	VIEWPOINT	
CCCS – Current-Controllled	F		Current Iprobe	IPROBE	

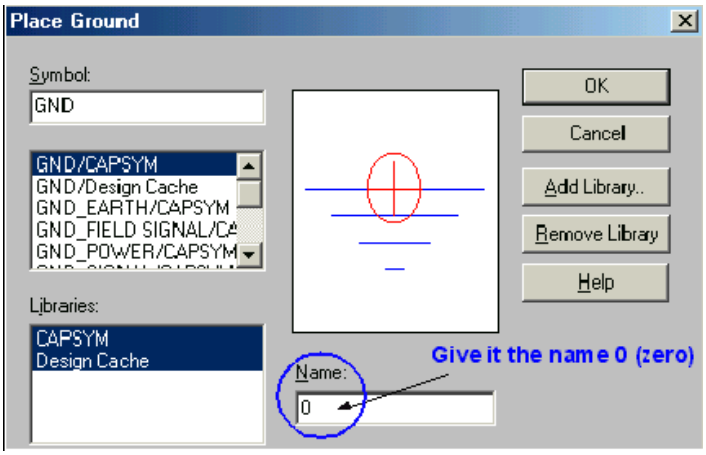
Current Source					
VCCS – Voltage-Controlled Current Source	G				
CCVS - Current-Controlled Voltage Source	H				

9. Place the resistors, capacitor and the DC voltage and current source.



- You can place the part on the Schematic window by the left mouse click.
- You can rotate the components by pressing the R key on the keyboard. In order to change a resistor from horizontal position to vertical, press the R key thrice. If the key is pressed once instead of thrice, the current through the resistor will be negative.
- To place another instance of the same part, click the left mouse button again.
- Hit the ESC key when done with a particular element.

10. After placing all parts, you need to place the Ground terminal by clicking on the GND icon (on the right side toolbar – see the figure under step 5). When the Place Ground window opens, select GND/CAPSYM and **give it the name 0 (i.e. zero).** Do not forget to change the name to 0, otherwise PSPICE will give an error or "Floating Node".



11. Now connect the elements using the Add Wire icon.

12. You can assign names to nets or nodes using the Add node name (aka Place net alias) button. We will do this for the output node and input node. Name these Out and In.

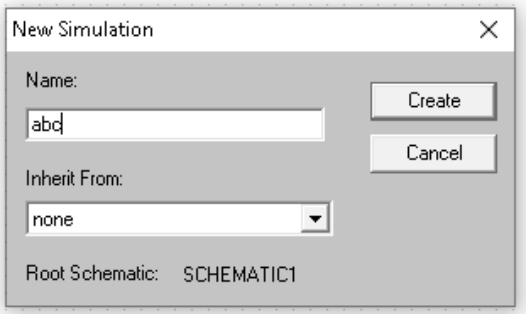
13. Change the values of the resistors by double-clicking on the number (1k) next to the resistor. You can also change the name of the resistor. Do the same for the capacitor and voltage and current source.

14. Save the project.

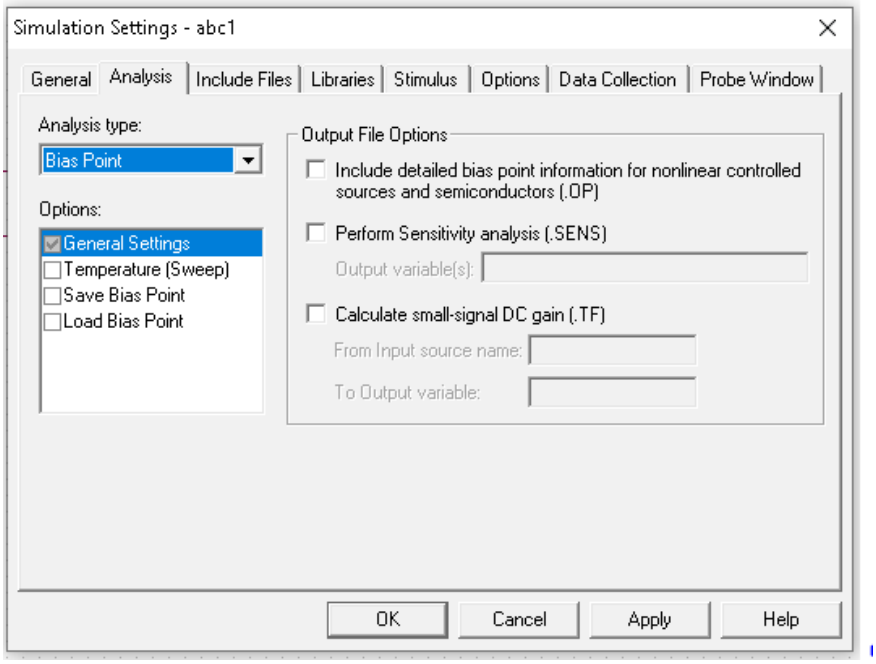
Simulation: DC analysis

1. With the schematic open, go to the PSPICE menu and choose NEW SIMULATION PROFILE.

2. In the Name text box, type a name. From the Inherit From List: select none and click Create.



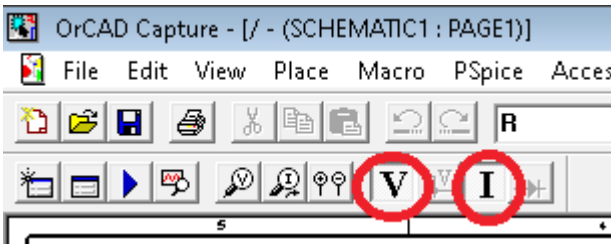
3. When the Simulation Setting window opens, for the Analyis Type, choose Bias Point and click OK.

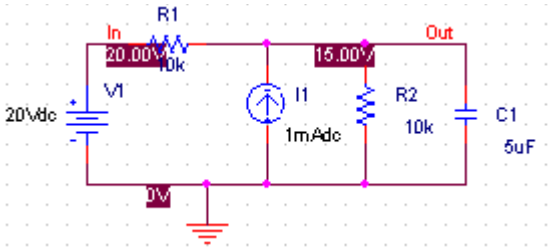


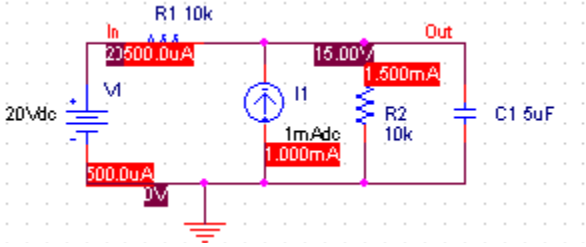
4. Now you are ready to run the simulation: select PSPICE → RUN from the menu at the top.

5. A window will open, letting you know if the simulation was successful.

6. To see the result of the DC bias point simulation, click on the V icon (Enable Bias Voltage Display) and I icon (current display) to show the voltage and currents.





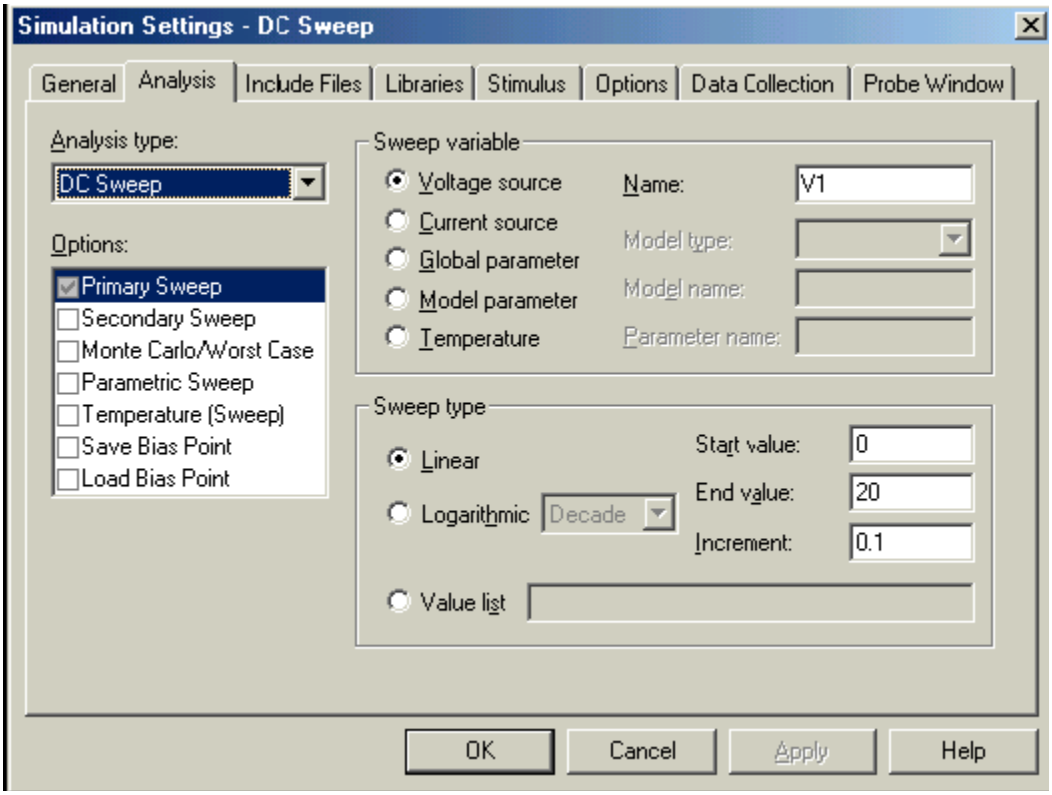


Simulation: DC Sweep

We will be using the same circuit but will evaluate the effect of sweeping the voltage source between 0 and 20V. We'll keep the current source constant at 1mA.

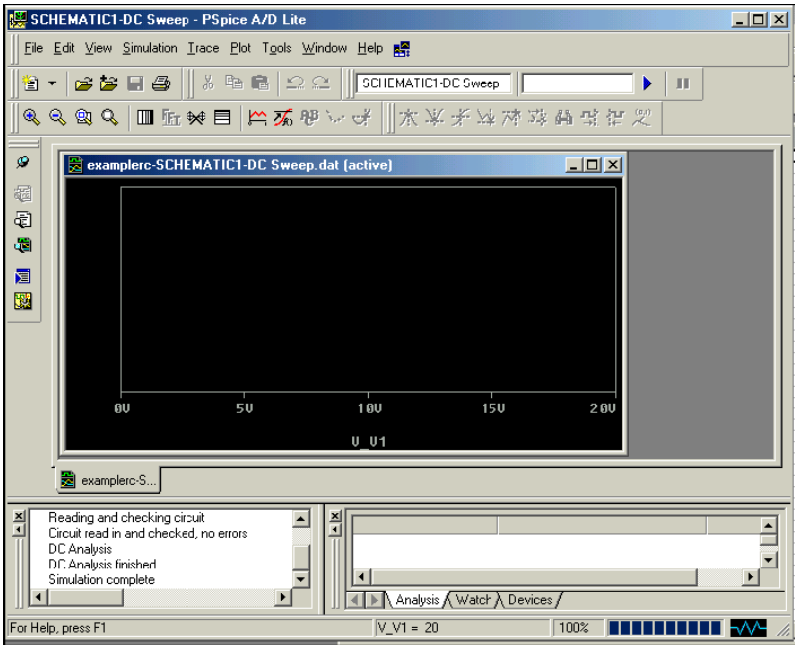
1. Create a new New Simulation Profile (from the PSpice Menu)

2. For analysis select DC Sweep; enter the name of the voltage source to be swept: V1. The start and end values are 0 V and 20 V respectively. Set the step size as 0.1V.

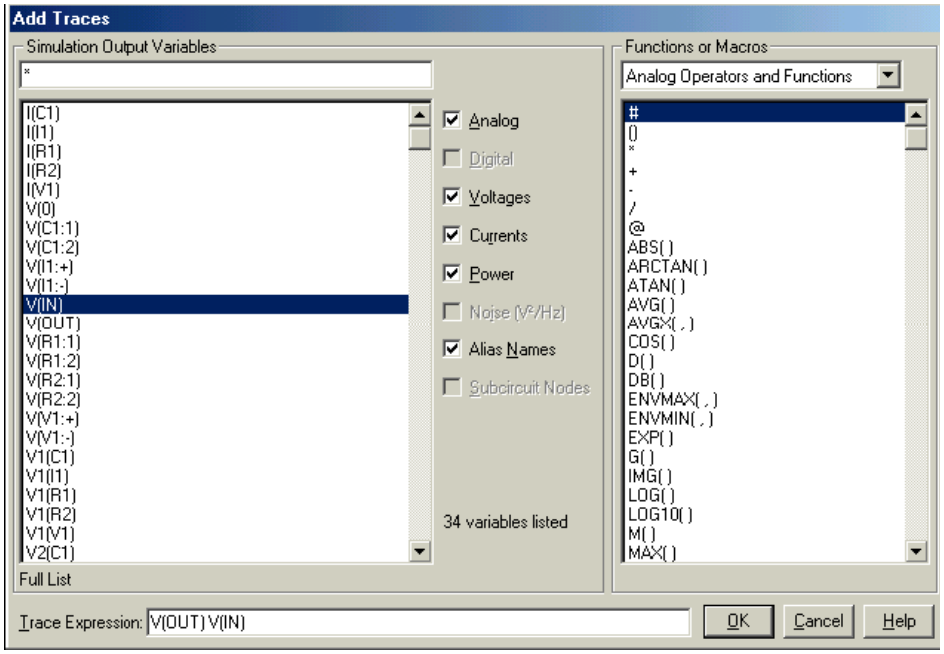


3. Run the simulation.

4. Once the simulation is finished a Probe window will open.



5. From the TRACE menu select ADD TRACE and select the voltages and current you like to display. In our case we'll add V(OUT) and V(IN). Click OK.



6. You can add a second Y Axis and use this to display e.g. the current in Resistor R2. Go to PLOT/Add Y Axis. Next, add the trace for I(R2).

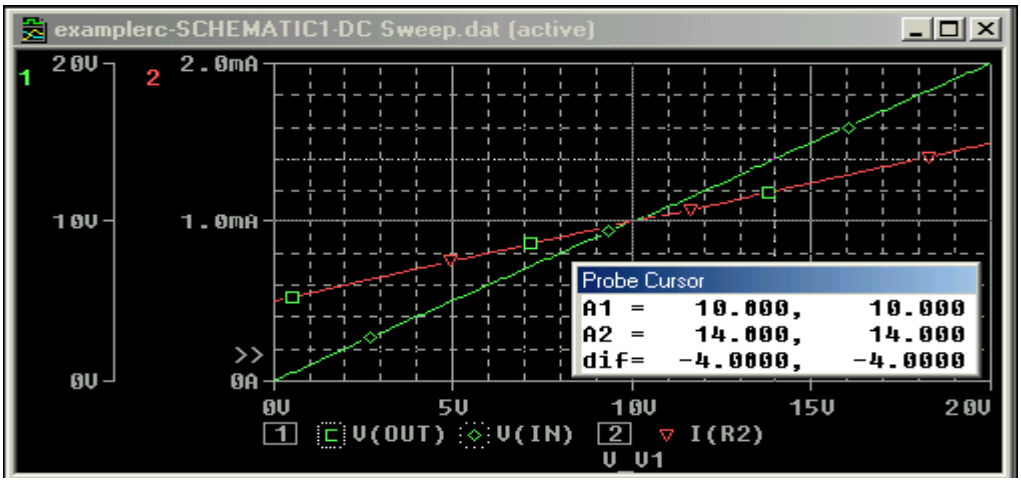
7. You can also use the cursors on the graphs for Vout and Vin to display the actual values at certain points. Go to TRACE/CURSORS/DISPLAY.

8. The cursors will be associated with the first trace, as indicated by the small rectangle around the legend for V(OUT) at the bottom of the window. Left click on the first trace. The values of the x and y axes are displayed in

the Probe window. When you right click on V(OUT) the value of the second cursor will be given together with the difference between the first and second cursor.

9. To place the second cursor on the second trace (for V(IN)), right click the legend for V(IN). You'll notice the outline around V(IN) at the bottom of the window. When you right click the second trace the cursor will snap to it. The values of the first and second cursor will be shown in Probe window.

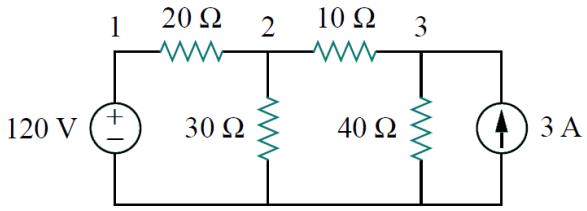
10. You can change the X and Y axes by double clicking on them.



Exercises

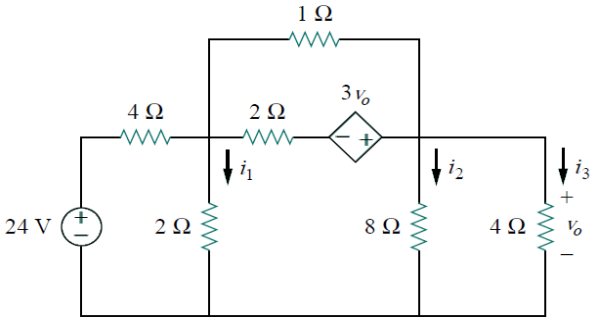
Exercise 1

Use PSPICE to find the value of voltage at each node in the circuit shown below. Verify the results theoretically.



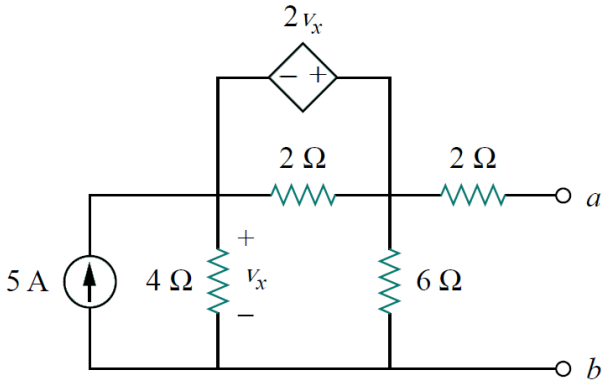
Exercise 2

Use PSPICE to find the values of currents i_1, i_2 and i_3 in the circuit shown below. Verify the results theoretically.
N.B: The connection for dependent sources in PSPICE follows their theoretical description. If needed, please see the tutorial available here - https://www.youtube.com/watch?v=WAIUHVXPdxM&ab_channel=SamsTutorials
If "GAIN" value of the dependent source is not shown by default on your schematic, double click on the dependent source to open its property window and set the gain from there. To return to the schematic, use the "Window" menu near the top.



Exercise 3

For the circuit shown below, use PSPICE to find the Thevenin and Norton equivalent circuits. Verify the results theoretically.
Hint: You can connect a test source between terminals a and b and sweep its value. Then from the I-V graph obtained across the test source, the Thevenin/Norton equivalent can be calculated.



Write-Up Requirements

A good laboratory report should contain a brief description of what the experiment was about, including circuit diagrams, and what you did, your data, your results, and anything else called for in the assignment, such as questions inserted in the laboratory. Answers to these questions require observations that need to be made at the time you do the experiment. The laboratory report should be written using the IEEE paper style consisting of a double-column single-space format, and should contain the following:

1. Title page - Title of the assignment/project, authors' name, and course name.
2. Abstract - Abstract of the assignment/project report.
3. Introduction
4. Main body of the assignment/project report including figures and explanations.
5. Conclusions
6. References (if needed)
7. Appendices (if needed)

For Laboratory no.1, the report should include screenshots of all the exrecise circuits built in PSPICE, screenshots of data reported from PSPICE, graphs from PSPICE and theoretical verification of the results obtained.

END